

Orbital Mechanics Golf

Demo 7 of 16 · Part four, Example projects · 20 minutes for stages 1–3, plus the Sandbox

You need: `index.html` open in any modern browser, on a projector or on each student's phone. No network, no install, no accounts. Nothing is fetched at run time and nothing leaves the machine. Chromebook-friendly.

Learning goals

One per stage, in the words the learner reads on the Takeaway card. There is nothing else to state and nothing to read out: these are what the stage demonstrates, and the check-yourself item at the foot of each stage tests exactly them.

- **Burn at apoapsis.** A prograde burn changes the orbit half a revolution away: burn at apoapsis and periapsis rises, burn at periapsis and apoapsis rises.
- **Escape at periapsis.** The same Δv buys energy in proportion to how fast you already are, so escape is cheapest at periapsis: 661 m/s here against 1,835 m/s from apoapsis. That is the Oberth effect.
- **Finite thrust.** Spread over time, a burn is cheapest centred on periapsis — 775 m/s centred against 919 m/s lit at periapsis — and with a weak engine six short kicks cost 731 m/s where one long burn costs 1,498.
- **Sandbox.** Every strategy that works, works for one reason: it spends Δv where v is already large.

Where this sits on the path

The first of the ten AI-built classroom tools in Part four, and the first of the two that are games. It carries no handover from Demo 6: what Part four has in common is not a chain of ideas but a discipline — each tool starts from a learning goal, exposes its assumptions, and makes its evidence available for challenge. This one's engine is a real gravity integrator, which is why the central result can be measured rather than asserted.

One question for the session

Ask it before anyone touches a control, and come back to it at the debrief.

"Escaping Earth needs a fixed amount of energy. You have a fixed tank. So does it matter where in the orbit you spend it?"

Most of the room will say no, and the arithmetic they have — Δv is Δv — supports them. Stage 2 is 20 minutes of them finding out that the question was about energy, not about speed.

Running it

Open ⚙ **Settings** → **Presentation mode** first: it scales the page for the back row and puts a 📝 **Notes** button in the header. The **Guide** card opens on load and explains the controls; dismiss it with *Start with Burn at apoapsis*.

- **0–3 min** Stage 1 on the projector together. Take the room's prediction by show of hands before you commit one on screen. Fly Level 1: → **Ap**, 🔥 **Plan burn**, drag Δv , read the dotted preview, commit. Vocabulary as you go: periapsis (low, fast), apoapsis (high, slow), prograde.
- **3–8 min** Students take stages 1 in pairs, Levels 1 and 2. Point them at the **Physics HUD**'s bottom row — "energy gained per 1 Δv " — and ask them to watch it as the ship falls.
- **8–16 min** The main event: stage 2, **Level 3 "Escape Artist"**. The stage will not hand over the flight view until a prediction is committed; that is deliberate, and it is worth saying so out loud. Par is 791 m/s and the tank holds 2,057, so the wasteful route still works — it just costs. When someone makes par, have them say *where* they burned and what the HUD read at the time.

- **16–19 min** Debrief on the projector from a student's energy-vs-time plot and the **show the math** panel under it, which evaluates $\Delta\epsilon = v \cdot \Delta v + \frac{1}{2}\Delta v^2$ on that student's own numbers and checks it against the simulation. The burn is a vertical jump in ϵ , and it is taller when v was larger for the *same* Δv .
- **19–20 min** Post the leaderboard. Stage 3 unlocks once a student has escaped, and Levels 6–7 make a good second session or homework.

Stage by stage

Stage 1 — Burn at apoapsis (Levels 1–2)

Question

You are at the top of an ellipse and you push forward. Which part of the orbit moves?

Predict prompt

Burn forward at apoapsis. Does periapsis rise, does apoapsis rise, or do both? Three buttons, captured, echoed beside the result.

Activity

Level 1 circularizes an ellipse; Level 2 raises apoapsis into a band. The two together are the stage, because between them they show the rule in both directions.

Expected observation

The dotted preview lifts the *far* side of the orbit while the near side barely moves. The observation cue fires at each apsis with the speed there — around 1.4 km/s at apoapsis on Level 3's orbit — and names it as the minimum energy-per- Δv point.

Check-yourself answer key

Because it raises the point half an orbit away, which from apoapsis is periapsis. The two wrong options are "apoapsis is where the engine is most efficient" (the refutation item; the opposite is true and stage 2 measures it) and "a single burn reshapes the whole orbit at once".

Teaching limit

Do not let this stage become the lesson. It is vocabulary and geometry; nothing about cost has been established yet, and a student who leaves stage 1 thinking apoapsis is "the good place to burn" has acquired the misconception stage 2 exists to break.

Stage 2 — Escape at periapsis (Levels 3–5)

Question

Escaping Earth needs a fixed amount of energy — so does it cost the same Δv wherever you burn?

Predict prompt

Level 3 starts on an ellipse with periapsis 11,100 km and apoapsis 70,100 km above the surface. Which burn point escapes for less Δv ? **The flight view does not appear until this is answered.**

Activity

→ **Pe** to fall to periapsis,  **Plan burn**, push Δv until the **ORBIT** chip flips from BOUND to ESCAPING. Then have them undo and try it from apoapsis with the same tank, which is the comparison the stage is built on. Levels 4 and 5 are the same idea applied to a transfer and to a powered flyby.

Expected observation

661 m/s from periapsis; 1,835 m/s from apoapsis; a factor of 2.77. Both figures are minima found against this exact engine. The HUD's "energy gained per 1 Δv " reads 6.09 km/s at periapsis and 1.39 at apoapsis — the same numbers, seen as they happen.

Check-yourself answer key

Item 1: **because you are moving fastest there, and the energy added is $v \cdot \Delta v + \frac{1}{2}\Delta v^2$.** The refutation option is "because you are closer to Earth" — see Misconception 1. Item 2: **no — an unpowered assist borrows the moon's motion; Oberth is about when you spend fuel.** See Misconception 2.

Teaching limit

$2.77\times$ belongs to *this* ellipse. Round the orbit out and the gap shrinks; on a circle there is no periapsis to prefer. What generalises is the direction of the effect, not its size, and the Details drawer says so in as many words.

Stage 3 — Finite thrust (Levels 6–7)

Question

Your engine needs hours to deliver its Δv . Where in the orbit should that burn happen?

Predict prompt

The full burn lasts a noticeable arc. Light it exactly at periapsis, or start before periapsis?

Activity

Levels 1–5 use impulsive burns; 6 and 7 give the engine 0.055 and 0.013 m/s². Point students at three readouts and they can compute an ignition point instead of guessing: the HUD's **Time to periapsis**, the planner's **Burn duration**, and the **lead** box beside $\rightarrow \mathbf{Pe}$, which stops the warp that many minutes short of the apsis. This stage is locked until a student has escaped in stage 2.

Expected observation

Level 6: 775 m/s with the burn centred on periapsis, 919 m/s lit at periapsis, against an impulsive ideal of 661 — so finite thrust always costs something, and timing decides how much. Level 7's engine is weak enough that one long burn smears around most of the orbit and costs 1,498 m/s, where six kicks of roughly 2.8 hours each cost 731.

Check-yourself answer key

Because all of it happens after the fast point, at falling speed. The refutation option is "a longer burn wastes more fuel as heat" — see Misconception 4.

Teaching limit

Steering losses are ignored entirely, and there is no rocket equation, so the finite-thrust penalty here has the right shape and not the right size. Treat 775-against-919 as "centring matters, and roughly this much".

Stage 4 — Sandbox (outside the path)

No goal, no par, 200 units of Δv , all three engines selectable, and a second fictional moon. Nothing is scored or saved. Its Predict card asks for a number rather than a button: how cheaply do you think you can escape? Write it down, then beat it. Its check-yourself item is the synthesis one — which single quantity decided every answer across the three stages — and the answer is your speed at the instant you spend the Δv .

The physics

An impulsive burn changes speed by Δv along the velocity, so kinetic energy per unit mass goes from $\frac{1}{2}v^2$ to $\frac{1}{2}(v+\Delta v)^2$:

$$\Delta \epsilon = \mathbf{v} \cdot \Delta \mathbf{v} + \frac{1}{2}\Delta v^2$$

The $\mathbf{v} \cdot \Delta \mathbf{v}$ term dominates, and it carries your speed at the instant of the burn. Gravitational potential is unchanged at that instant — you have not moved — so the whole difference between a cheap burn and an expensive one is the v in that term. On an ellipse, speed peaks at periapsis.

The HUD's "energy gained per 1 Δv " readout is therefore numerically just your current speed, and it is worth asking a calculus class to differentiate $\frac{1}{2}v^2$ and see why.

Where does the extra energy come from? The propellant's chemical energy is fixed. What changes is how it divides between the ship and the exhaust: at high speed the engine acts over more distance per second, so more of that energy ends up in the ship and less in the plume left behind.

Misconceptions to catch

Misconception 1: "The burn is better at periapsis because you are *closer* to the planet." Proximity is not the mechanism — **speed** is. $\Delta \epsilon = \mathbf{v} \cdot \Delta \mathbf{v} + \frac{1}{2}\Delta v^2$ contains v , not r . Being deep in the well matters only because falling there is what made the ship fast; a ship moving equally fast far from the planet would gain just as much from the same burn. This is the wrong option on stage 2's first check item.

Misconception 2: "The Oberth effect is the same as a gravity assist." An **unpowered assist** (Level 5's moon) takes energy from the *moon's orbital motion* — no fuel, and the moon slows imperceptibly in exchange. An **Oberth burn** spends fuel and merely spends it at the most profitable moment. A powered flyby combines both, which is what Level 5 actually is.

Misconception 3: " Δv is like distance — it costs the same anywhere." The Δv *spent* is the same anywhere; the *energy it buys* is not. That is the whole game, and it is the third wrong option in stage 2.

Misconception 4: "A long burn costs more because the engine wastes fuel." The engine never changes within a level. What changes is the speed at which each second of thrust is delivered. Stage 3's refutation item is exactly this.

Misconception 5: "A very tall ellipse has escaped." It has not: escape is $\epsilon > 0$, not a distance. A student who crosses the system edge while still bound gets told so on screen, with their actual ϵ quoted, because the geometry is genuinely misleading.

Discussion questions

1. Level 3's energy plot shows a taller jump in ϵ for the same Δv at periapsis. Which term of $\Delta \epsilon = \mathbf{v} \cdot \Delta \mathbf{v} + \frac{1}{2}\Delta v^2$ explains this, and what would have to be true for the other term to matter more?
2. The HUD says "energy gained per 1 Δv = your current speed". Differentiate $\frac{1}{2}v^2$ and show why.
3. Why does a prograde burn at periapsis raise the *apoapsis*, on the far side of the orbit?
4. In Level 5, where does the escape energy come from when you fly close behind the moon without burning? Name the thing that slows down in exchange.

5. A mission that wants to be captured into orbit has to lose energy. Where on the approach should it burn, and does the same argument run backwards?
6. Could you escape with a radial-out burn? Try it, then explain from the formula why pushing "straight up" wastes fuel.
7. Level 7's six kicks cost half what one long burn costs. What does that suggest about a spacecraft whose engine is even weaker — and what eventually limits how far you can push the idea?

Numbers, and where each one comes from

Every figure in this guide and in the demo is one of three things, and the demo labels each of them on its face.

Quantity	Value	Provenance
Earth μ , radius	$398,600 \text{ km}^3/\text{s}^2 \cdot 6,371 \text{ km}$	Sourced. NASA NSSDC Earth Fact Sheet, read 2026-09-16: "GM ($\times 10^6 \text{ km}^3/\text{s}^2$) 0.39860", "Volumetric mean radius (km) 6371.000".
Level pars	L1 712 · L2 1,107 · L3 791 · L4 1,819 · L5 1,424 · L6 838 · L7 823 m/s	Measured against this exact engine. The integrator has no random element, so there is no seed: re-running test-physics.js reproduces every figure.
L3 escape minima	661 m/s at periapsis · 1,835 at apoapsis	Measured by bisection; the periapsis figure also matches closed-form $v_{\text{esc}} - v_p$ to three figures.
L6 / L7 finite thrust	775 centred · 919 lit at Pe · 731 in six kicks · 1,498 in one burn	Measured , same harness.
The moon	invented orbit, mass and size	Illustrative. Not our Moon — closer and heavier relative to this setup, so an assist fits a class period. Labelled on the map itself.
Why low-thrust stages use repeated perigee burns	—	Reasoned from the 731-against-1,498 measurement. We did not source a specific flown mission's burn plan; present it as the principle, not as a citation.

Do not quote these numbers as spaceflight figures. The model has no atmosphere, no mass loss and so no rocket equation, no steering cost and no third-body perturbations beyond the one fictional moon. It is built to isolate one relationship, and it strips away everything that would compete with it.

Presenter controls

- **⚙ Settings** holds the three presenter controls in order — **Open Presenter Notes** (this page, on screen), **Presentation mode** (larger type plus a header Notes button) and **Reset**.
- **Reset** puts the lesson back to a first load in place, with no page reload: Level 1 on the pad, an empty Δv tally, the trajectory and undo stack cleared, warp back to 1×, the HUD open, all three predictions forgotten, stage 3 locked again, and the mission badges and saved name cleared. It deliberately leaves two things alone: **presentation mode**, because re-shrinking the projector between rooms would be hostile, and the saved **class leaderboards**, because those hold other students' entries. Use *Clear scores* for those, which asks first.
- **Keys:** Space pause · B burn planner · Enter commit · Esc cancel · Z undo · X cut engine · , . warp down/up · P / A warp to Pe / Ap · H HUD · F follow camera · M briefing · R restart level · ? Guide · N Presenter Notes.

- **Undo** rewinds the last burn and refunds the fuel, and it is offered again after a crash. Students should iterate, not restart.
- The four stage tabs jump between the lesson's beats; picking a level from  **Levels** moves the tabs with it. A `#stage-3` link in a lesson plan will clamp to the furthest unlocked stage rather than routing around the gate.

Orbital Mechanics Golf · Bryant Harrison · Murray State University · Offline teaching game. Open `index.html` to play. Physics: velocity-Verlet coasts with RK4 powered arcs; all level pars are verified against the shipped physics by `test-physics.js` and the built page by `test-bundle.js`. `src/demo-guide.html` is the canonical source of this document; the shipped `teacher-guide.html`, the PDF beside it, and the in-app presenter notes are all generated from it at build time.